

Grace Bianchi

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EDUCATION

University of California, Santa Barbara

B.S. Statistics and Data Science, B.A. Economics

Expected Graduation: June 2027

3.88/4.00 GPA

Honors Thesis in Economics, Data Science Capstone – Admitted, Fall 2026

Relevant Coursework: Stochastic Processes, Probability Theory, Game Theory, Linear Algebra, Mathematical Statistics, Machine Learning, Mathematical Finance, Regression Analysis, Econometrics, Vector Calculus, Discrete Math

PROFESSIONAL EXPERIENCE

MUFG | Data Quality and Solutions Summer Analyst

June 2026 – August 2026

- Extracted 10 years of loss-run data from PDFs using Python, then cleaned and structured claims data in PostgreSQL
- Modeled claim frequency with Poisson and severity with Pareto distributions, using Monte Carlo simulation to estimate umbrella layer exceedance probabilities and support a coverage reduction, saving \$250K in premiums
- Validated reverse-engineered financial crimes rules in SQL during a platform migration for legacy-to-new parity

SageView Advisory Group | Data and Analytics Intern

June 2025 – August 2025

- Built a client segmentation model in R using weighted z-scores to score and tier 150+ institutional clients, and applied a decision tree to generate targeted improvement recommendations based on lowest-performing metrics
- Developed a SQL and Power BI dashboard to search and visualize nested account, plan, and contact data
- Prepared investment reports for client meetings, compiling portfolio holdings and YTD returns

University of California, Irvine Spatial Neuroscience Lab | Research Assistant

June 2022 – January 2023

- Developed Python scripts to process VR-based spatial navigation experimental data for statistical analysis
 - Supported integrating VR, fMRI, and diffusion-weighted imaging to study neural systems underlying navigation
 - Assisted in EDA and scientific literature review on hippocampal and cortical contributions to spatial cognition
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PROJECTS

SPY Markov Regime-Switching Volatility Forecaster | Python, arch, hmmlearn, scikit-learn

- Benchmarked EWMA, GARCH (1,1), and HAR-RV forecasts on a 2015-2022/2023+ split; HAR-RV achieved the lowest RMSE (4.90 vs 7.39-8.70), confirmed by a linear regression model converging on the same signal
- Built a 4-state HMM (Calm/Normal/Stressed/Panic) and a regime-conditional forecaster on top; a Diebold-Mariano test showed the regime edge was directionally present but not statistically significant on point forecasts
- Replaced normal-tail VaR/ES with a Generalized Pareto tail fit, passing Kupiec/Christoffersen backtests at 99%/95%, confirming tail modeling fixed calibration, then deployed as a live GitHub Actions forecaster

Stellar Classification: Using Machine Learning to Assess the Role of Redshift | R, tidymodels, xgboost, kkn

- Fit 14 classification models (7 types x full/reduced feature sets) in R to classify stellar objects from SDSS data
- Demonstrated that redshift is a critical predictor – removing it drops Random Forest test accuracy from 97.9% to 84.1%, with STAR classification most impacted (39% misclassification rate without redshift)
- Tuned hyperparameters via 10-fold cross-validation; evaluated best models using ROC AUC, accuracy, ROC curves, and confusion matrices

Housing Market Segmentation using Clustering and Association Rules | R, cluster, FactoMineR, arules

- Applied k-means, hierarchical (Ward.D2), and PAM clustering to 1,047 homes (k=3, validated via Hopkins statistic H=0.999); PCA confirmed a dominant size-value axis explaining 49.1% of variance across 7 housing features
 - Mined 6,749 Apriori association rules, finding that a large living area and 3+ bathrooms predict an expensive price tier with 100% confidence and a lift of 2.94
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SKILLS & ACTIVITIES

Programming Languages: Python, R, SQL, Java, Stata

Tools: PostgreSQL, Git, Jupyter, Power BI, Excel, RStudio, LaTeX

Awards and Honors: UCSB Dean's Honors, AP Scholar with Distinction, AMC Math Competition Participant

Activities: Undergraduate Learning Assistant: Intermediate Microeconomic Theory, Statistics for Economics, UCSB Data Science Club, UCSB Data Science Collaborative, Kappa Alpha Theta, Scitrek